Learning Which Route to Trust: Stability-Aware Bayesian Next-Hop Selection for Infrastructure-Independent Disaster Messaging

Pinyen Cheng
Taoyuan, Taiwan

When cellular sites lose power and backhaul after a disaster, short text messages must hop between phones, shelters, and regional gateways over intermittent, low-rate links. Choosing the next hop is then a prediction problem: which neighbor is most likely to deliver the message before it expires? We present STAR (Stability-aware TAR routing), a fully local next-hop policy for the Taiwan Resilient Communication Network that (a) predicts route stability Ψ, the probability of on-time delivery, from discounted Bayesian posteriors over link frame success and availability combined with loop-free, authenticated distance-vector advertisements; (b) decides with a priority-aware rule that is risk-averse and replicates distress messages, yet explores with Thompson sampling on non-urgent traffic; and (c) learns from outcomes through fragment acknowledgments, signed end-to-end receipts treated as falsifiable deadlines, and overheard forwarding, which together calibrate trust in each neighbor's route claims. In a discrete-event simulator with LoRa time-on-air and airtime-accounted control traffic, evaluated on 20 held-out seeds of a synthetic two-region disaster network, STAR delivered 97.7% of messages and all distress messages, against 89.6% for a loop-free ETX distance-vector baseline and 90.5% for PRoPHET, with 44% lower median latency than the former. After a simulated aftershock it delivered 92.8% versus 82.5% and 84.9%. Under route poisoning it retained several times the delivery of an unprotected distance-vector protocol, although replication-based protocols remained more robust. Gains vanish at saturation. All results are synthetic.

Keywords: delay-tolerant networking, learned routing, Thompson sampling, Bayesian link estimation, disaster communication, LoRa, route poisoning

Introduction

A disaster that cuts grid power silences cellular sites long before it damages them. After the magnitude 7.2 earthquake off Hualien on 3 April 2024, Taiwan's three major carriers reported 172 affected base stations, mostly because of interrupted power and transmission links (Central News Agency, 2024), and a highway collapse left the Tianxiang area without communication until satellite terminals were airlifted in three days later (Ministry of Digital Affairs, 2024). Backhaul can be restored at a few points; what remains missing is a path between *people, shelters, and coordinators* in the last few kilometers.

The Taiwan Resilient Communication Network (TRCN) is a research architecture for that gap: phones reach shelters over Wi-Fi Aware or Bluetooth, shelters and regional gateways form a LoRa mesh, and regions interconnect over whatever back-

Simulation code, raw results, and scripts that regenerate every table and figure are available at https://github.com/chengmatt416/trcn. This work has no affiliation with any government agency; no radio was operated and no personal data were collected. The author used generative AI assistance in this work (see the Use of Generative AI section). Correspondence concerning this article should be addressed to Pinyen Cheng, Taoyuan, Taiwan.

bone survives. Messages are short, end-to-end encrypted, and carried store-and-forward with per-hop custody and recipient-signed receipts. The earlier TRCN prototype routed with *preloaded* next hops and ignored control-plane airtime, which its own documentation lists as the first limitation to fix.

This paper addresses that limitation. Its research question is: *Using only information a node can observe locally, can a node predict which neighbor offers the most stable route to a destination, act on that prediction, and improve it from the outcomes of its own decisions?* Stability here has a precise meaning: the probability that a message is delivered before its deadline.

Two properties make the problem hard. First, links in a disaster are *intermittent and non-stationary*: a Wi-Fi link to a phone appears and disappears as people move or screens sleep; a LoRa link fades with interference; an aftershock removes links permanently. Classic shortest-path metrics such as ETX (De Couto et al., 2003) capture loss but not availability or deadlines; DTN utilities such as PRoPHET (Lindgren et al., 2003) capture encounter history but replicate aggressively, which LoRa's scarce airtime cannot afford (Bor et al., 2016). Second, feedback is *delayed and partial*: a node sees per-hop acknowledgments quickly, but whether a message reached its destination is known only to the source and only when a receipt returns, if it returns at all.

Contributions. (1) We formulate next-hop selection in TRCN as on-line prediction of a deadline-aware stability score Ψ and derive a tractable, fully local estimator that composes Bayesian link posteriors with loop-free, authenticated distance-vector summaries (see The STAR Model). (2) We give a priority-aware decision rule: risk-averse with bounded replication for distress traffic, Thompson-sampling exploration for non-urgent traffic, and explicit hold decisions for store-carry-forward (Algorithm 1). (3) We close the learning loop with three outcome channels (selective fragment acknowledgments, receipts treated as falsifiable delivery deadlines, and watchdog overhearing) and show that *trust as calibration* of neighbors' claims bounds route poisoning. (4) We release an open simulator with LoRa time-on-air, airtime-accounted control traffic, collisions, crashes, and blackhole attacks, and evaluate STAR against six baselines over 1,520 runs with held-out seeds (see Evaluation Method through Results).

Related Work

Delay-tolerant routing. DTN routing (Fall, 2003; Jain et al., 2004) spans replication (epidemic routing, spray-and-wait), utility-based forwarding (PRoPHET, MaxProp), resource-allocation formulations (RAPID), and contact-plan routing for scheduled links (Balasubramanian et al., 2007; Burgess et al., 2006; Fraire et al., 2021; Lindgren et al., 2003; Spyropoulos et al., 2005; Vahdat & Becker, 2000). Replication buys robustness with airtime and buffer space; contact-plan routing assumes a known schedule, which disasters do not provide. Simulation tools such as the ONE (Keränen et al., 2009) popularized evaluation under mobility traces. TRCN's shelter backbone is mostly static but intermittent, which places it between classic mesh routing and opportunistic DTN.

Link-quality-aware mesh routing. ETX and its time-based variants weight links by expected transmissions (De Couto et al., 2003); link estimators are surveyed by Baccour et al. (2012), and burst-error channels go back to the Gilbert model (Gilbert, 1960). Distance-vector protocols avoid loops with destination sequence numbers (Perkins & Bhagwat, 1994); Babel adds a feasibility condition that keeps routing loop-free even during convergence (Chroboczek & Schinazi, 2021). STAR inherits Babel's feasibility condition but replaces its additive metric with a probabilistic, deadline-aware score.

Learning-based routing. Q-routing learns per-destination delivery-time estimates from neighbor feedback (Boyan & Littman, 1994) and has predictive and dual variants (Choi & Yeung, 1996; Kumar & Miikkulainen, 1997). Reinforcement learning has been applied to DTN forwarding and replication (Dudukovich et al., 2017; Rolla & Curado, 2013); Bhavani et al. (2023) review machine-learning DTN routing, and graph neural networks model network performance in software-defined networks (Rusek et al., 2020). Bandit formulations

of routing study regret when per-link or end-to-end feedback is observed (Gai et al., 2012; Talebi et al., 2018); a key finding is that per-link feedback is far more valuable than end-to-end feedback, which motivates STAR's use of fragment-level acknowledgments. Thompson sampling (Russo et al., 2018; Thompson, 1933) and its discounted, non-stationary variants (Garivier & Moulines, 2011; Lattimore & Szepesvári, 2020; Raj & Kalyani, 2017) underlie STAR's exploration and forgetting. Unlike Q-routing, STAR predicts a *probability* of on-time delivery rather than an expected time, and it uses the uncertainty of its own estimates to set exploration and risk by priority.

Secure routing. Secure AODV authenticates sequence numbers with hash chains (Guerrero Zapata & Asokan, 2002); watchdogs detect neighbors that do not forward (Marti et al., 2000). STAR combines both with outcome-based calibration so that an insider that advertises impossible routes loses traffic quickly.

Disaster LoRa networks. LoRa's range comes with long time-on-air and limited scalability (Adelantado et al., 2017; Augustin et al., 2016; Bor et al., 2016). Community stacks such as Meshtastic and Reticulum (Meshtastic, 2026; Qvist, 2026) and recent field studies (Khamaisi et al., 2025) demonstrate off-grid messaging; surveys catalogue the routing challenges of disaster ad hoc networks (Reina et al., 2015). To our knowledge, none of these combines deadline-aware stability prediction, outcome-calibrated trust, and accounted control airtime.

System Model

Network. Nodes are phones, shelters, regional gateways, coordinators, and relays. A link $\ell = (i, j)$ uses a medium (Wi-Fi Aware, BLE, LoRa with spreading factor SF, Ethernet, or microwave) with a maximum frame size and a per-frame airtime $t_f(\ell)$. LoRa time-on-air follows the Semtech modem equation (Semtech Corporation, 2013):

$$t_f = \frac{2^{\mathrm{SF}}}{B}\Big[(n_{\mathrm{pre}} + 4.25) + 8 + \max\Big(\Big\lceil \frac{8L - 4\mathrm{SF} + 44 - 20H}{4(\mathrm{SF} - 2D_E)} \Big\rceil (c_r + 4),\ 0\Big)\Big], \tag{1}$$

for payload L bytes, bandwidth B, coding rate $4/(4 + c_r)$, header flag H and low-data-rate flag D_E. LoRa links in a region share one collision domain with probabilistic carrier sense.

Link dynamics. Each link alternates between up and down periods drawn from exponential distributions (a continuous-time on/off process) and, while up, loses each frame independently with probability q_ℓ. Scenarios may force outages, fail links permanently, or crash nodes, which loses their RAM custody.

Messages. A message m has source, destination mailbox d, priority $p \in \{0, 1, 2, 3\}$ (P0 = SOS, P1 = receipts, P2 = private, P3 = supply), wire size (payload plus 192 bytes of envelope and laboratory cryptographic overhead), and a lifetime T_m (900 s by default). It is fragmented into TDF frames with a 24-byte header. Each hop is an *exchange*: the sender transmits the missing fragments; the receiver returns an acknowledgment carrying a fragment bitmap and, once all fragments have arrived and the message is admitted, a custody decision. Up to four rounds are attempted before the exchange fails and the sender backs off. Custody is not delivery: the source counts a message as delivered only when the destination's signed RECEIPT arrives.

Objective and constraints. Each node i holding a copy of m must decide, using only local state, whether to forward it and to which neighbor j, so as to maximize

$$\Pr[m \text{ delivered before } t_e(m)], \tag{2}$$

with ties broken toward lower latency and higher priority traffic served first. No node has the topology, link states, or other nodes' queues; all control information must be carried in airtime-accounted beacons or acknowledgment piggybacks.

The STAR Model

Figure 1 summarizes STAR. Each node keeps four estimators fed by local evidence, combines them into a stability score per neighbor, and applies a priority-dependent decision rule. Table A1 in the Appendix lists all parameters.

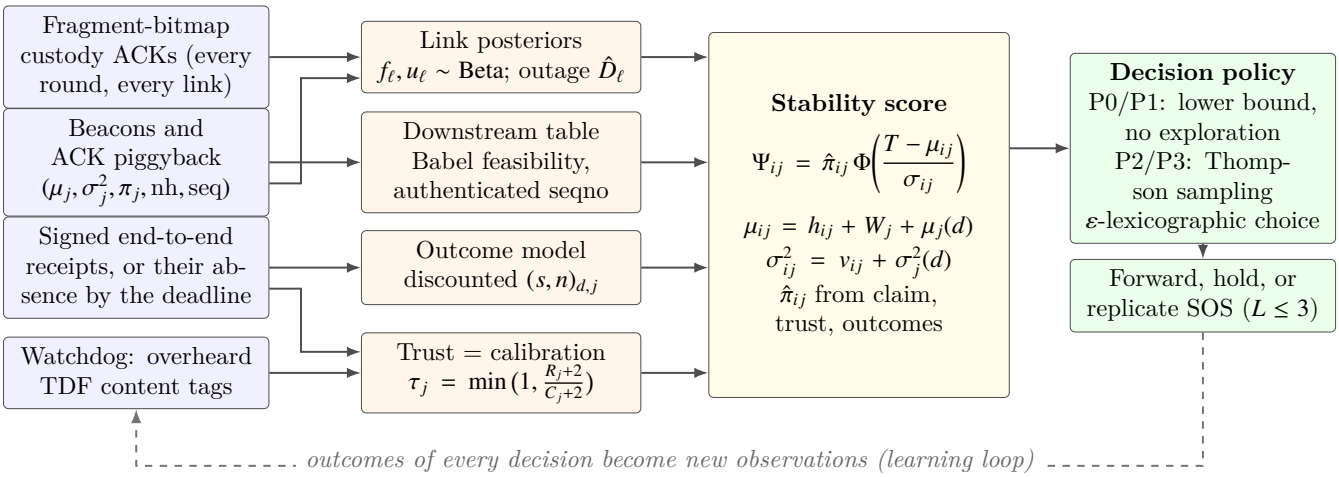


Figure 1

The STAR model at one node. Four local evidence streams feed four estimators; their outputs combine into the stability score Ψ, which a priority-aware policy turns into a forwarding decision. The outcome of every decision is observed again, closing the learning loop. No component requires global topology knowledge.

Link posteriors

For each incident link ℓ, node i keeps two Beta posteriors: frame success $f_\ell \sim \text{Beta}(\alpha_f, \beta_f)$ and availability $u_\ell \sim \text{Beta}(\alpha_u, \beta_u)$. A fragment-bitmap acknowledgment reporting k of n fragments received adds $(k, n-k)$ to (α_f, β_f). A received beacon, or any acknowledgment, adds one success to u_ℓ; each missed beacon (detected by a sequence gap) or unacknowledged round adds a failure. To track non-stationarity, before each update every pseudocount is discounted toward its prior,

$$\alpha \leftarrow \alpha_0 + (\alpha - \alpha_0)\, 2^{-\Delta t/H}, \tag{3}$$

with half-lives $H_f = 900\,\text{s}$ and $H_u = 240\,\text{s}$, which is the time-based analogue of discounted Thompson sampling (Raj & Kalyani, 2017). Node i also learns the mean outage length $\hat{D}_\ell$ as an exponentially weighted average of observed silent gaps. Because a long silence is itself evidence, the current availability belief is $\tilde{u}_\ell = \bar{u}_\ell \exp(-(s_\ell - 1.6B)^+/\hat{D}_\ell)$, where s_ℓ is the time since the last evidence that ℓ was up and B is the beacon period.

Hop model

For a message of n_ℓ fragments, frame time t_f, and acknowledgment time t_a, the hop delay has mean and variance

$$\begin{aligned} h_{ij} &= \frac{n_\ell t_f}{f_\ell} + t_a(1 + 3(1 - f_\ell)) + (1 - \tilde{u}_\ell)\hat{D}_\ell + W_i, \\ v_{ij} &= \frac{n_\ell(1 - f_\ell)}{f_\ell^2} t_f^2 + (1 - \tilde{u}_\ell)\hat{D}_\ell^2 + 1, \end{aligned} \tag{4}$$

where the first terms are the geometric retransmission cost under selective repeat, the third is the expected wait for a down link to return (the exponential-outage assumption makes the residual outage memoryless), and W_i is the node's own high-priority backlog. The value of f_ℓ depends on the decision mode: the posterior mean, a lower credible bound $\bar{f} - 1.28\,\text{sd}(f)$, or a Thompson sample.

Downstream summaries and loop freedom

Each node advertises, for a rotating subset of destinations, a compact tuple

$$(\mu_i(d),\ \sigma_i^2(d),\ \pi_i(d),\ \text{next hop},\ \text{seq}_d)$$

in 11 bytes, plus its queue backlog. The same tuple is piggybacked on every custody acknowledgment for the message's destination, so active routes are refreshed at no extra transmission. Loops are prevented with Babel's feasibility condition (Chroboczek & Schinazi, 2021): node i keeps a feasible distance $\text{FD}_i(d) = (s, \mu)$ and accepts neighbor

j's route only if $\mathrm{seq}_j > s$, or $\mathrm{seq}_j = s$ and $\mu_j(d) < \mu$; entries whose advertised next hop is i itself are ignored (poisoned reverse). Each destination increments its own sequence number every four beacons, and every number is accompanied, when it changes, by an 8-byte hash-chain authenticator in the style of SAODV (Guerrero Zapata & Asokan, 2002), so that no relay can advertise freshness the destination has not released.

Given neighbor j's entry of age a, with $a^+ = \max(0, a - 2B)$ the age beyond the expected refresh, the composed route through j has

$$\mu_{ij} = h_{ij} + W_j + \mu_j(d), \qquad \sigma_{ij}^2 = v_{ij} + \sigma_j^2(d) + (0.2a^+)^2, \tag{5}$$

and a claimed not-lost probability $c_{ij} = \pi_j(d)\, 2^{-a^+/H_s}$.

Reliability posterior and trust

Claims are evidence, not facts. Node i keeps, for each neighbor, a trust value equal to the *calibration* of that neighbor's past promises,

$$\tau_j = \min\Big(1,\ \frac{R_j + 2}{C_j + 2}\Big), \tag{6}$$

where C_j accumulates the delivery probability j claimed for each message handed to it and R_j the deliveries actually observed (see Learning from outcomes). For each (d, j) the source also keeps discounted counts (s_{dj}, n_{dj}) of end-to-end outcomes. The not-lost probability is the posterior mean

$$\hat{\pi}_{ij} = \frac{K\tau_j c_{ij} + s_{dj}}{K + n_{dj}}, \tag{7}$$

with $K = 4$ pseudo-observations for the calibrated claim. Neighbors with $\tau_j < 0.4$ are excluded; on exclusion, every route and feasible distance learned through j is discarded so that poisoned sequence state cannot starve honest routes.

Stability score

Approximating the remaining delay as Gaussian with mean μ_{ij} and variance σ_{ij}^2, a sum of many independent hop components, the predicted probability that a message with remaining lifetime T is delivered on time through j is

$$\Psi_{ij} = \hat{\pi}_{ij}\, \Phi\!\left(\frac{T - \mu_{ij}}{\sigma_{ij}}\right), \tag{8}$$

where Φ is the standard normal distribution function. Ψ couples the three ingredients that shortest-path metrics separate: how often a route loses messages ($\hat{\pi}$), how long it takes (μ), and how predictable it is (σ). For a message close to expiry, a fast but erratic route can lose to a slower, steadier one; for a fresh message the difference vanishes and delay decides.

Algorithm 1 STAR next-hop decision at node i for a message copy m

Require: copy m (destination d, priority p, expiry t_e, path memory $\mathcal{P}$, tokens L_m)

$T \leftarrow t_e - t_{\text{now}}$; **if** $T \le 0$ **then return** DROP
mode $\leftarrow$ SAMPLE if $p \ge 2$ else LOWERBOUND
for all neighbours $j \notin \mathcal{P}$ with a feasible, authenticated route to d and $\tau_j \ge \tau_{\min}$ **do**
    draw or bound f_ℓ, u_ℓ according to mode; compute (h_{ij}, v_{ij}) ▷ Eq. 4
    $\mu_{ij}, \sigma_{ij}^2, \hat{\pi}_{ij}$ from downstream table and outcome model ▷ Eqs. 5–7
    $\Psi_{ij} \leftarrow \hat{\pi}_{ij}\, \Phi((T - \mu_{ij})/\sigma_{ij})$
$C \leftarrow \{j : \Psi_{ij} \ge \max_k \Psi_{ik} - \varepsilon_p\}$; $j^\star \leftarrow \arg\min_{j \in C} \mu_{ij}$
if $\tilde{u}_{\ell(j^\star)} < 0.3$ and no alternative within $\Psi_{ij^\star}/2$ is believed up **then**
    return HOLD ▷ store-carry-forward
if $p = 0$, i is the source and $\Psi_{ij^\star} < \Psi^*$ **then**
    $L_m \leftarrow \min(L_{\max}, \lceil \ln(1 - \Psi^*)/\ln(1 - \Psi_{ij^\star}) \rceil)$ ▷ SOS replication
return FORWARD$(j^\star,\ \max(1, \lfloor L_m/2 \rfloor))$

Decision policy

Algorithm 1 turns scores into actions. **Risk by priority.** Distress messages and receipts (P0, P1) are scored with lower credible bounds and are never used to explore; private and supply messages (P2, P3) are scored with Thompson samples of f_ℓ and u_ℓ (Russo et al., 2018), so uncertain links are tried in proportion to the probability that they are best.

ε-lexicographic choice. Among neighbors whose Ψ is within ε_p of the best ($\varepsilon = 0.03$, or 0.01 for P0), STAR picks the lowest expected delay, which prevents small, noisy differences in $\hat{\pi}$ from selecting long detours. **Hold.** If the preferred link is believed down ($\tilde{u} < 0.3$) and no comparable alternative is up, the message is held rather than transmitted into a dead link, which saves airtime. **SOS replication.** When the best single next hop of a P0 message at its source has $\Psi < \Psi^* = 0.95$, STAR sets $L = \min(3, \lceil \ln(1-\Psi^*)/\ln(1-\Psi) \rceil)$ copy tokens, the number of independent routes of quality Ψ needed to reach Ψ^*, and spreads them over distinct next hops.

Learning from outcomes

STAR learns from three outcome channels at different time scales.

Per round (seconds). Fragment bitmaps update f_ℓ; acknowledgments and their absence update u_ℓ and $\hat{D}_\ell$; piggybacked tuples refresh downstream summaries.

Per message at the source (minutes). A route claim is treated as a *falsifiable commitment*. When the source hands a message to j, it schedules a check at

$$t + \mathrm{clip}(2(\mu_j(d) + 2\sigma_j(d)) + 60,\ 120,\ T_m + 300)\ \text{s}. \quad (9)$$

If the signed receipt has not arrived by then, the source records a provisional failure ($n_{dj}{+}1$, $C_j{+}c$); a late receipt converts it to a success ($s_{dj}{+}1$, $R_j{+}1$). A neighbor that promises fast delivery is therefore tested quickly, and one that lies about delay loses trust within minutes.

Per hop by overhearing. TDF frames carry the same 8-byte content tag on every hop. When i hands a message to j whose advertised next hop shares i's LoRa channel, i expects to overhear j forwarding that tag before a deadline derived from j's own claimed delay and backlog. Overhearing adds $(1, 1)$ to (R_j, C_j); silence adds $(0, 1)$. This watchdog (Marti et al., 2000) gives relays, not only sources, evidence about their neighbors.

Overhead and complexity

A STAR advertisement entry costs 11 bytes, plus 8 bytes when its sequence number changes; acknowledgments carry 8 more bytes. Beacons rotate 24 entries per period ($B = 60$ s) so that a beacon fits in one 222-byte LoRa frame. Computing Ψ for one neighbor is $O(1)$; a decision is $O(\deg i)$; state is $O(|\mathcal{D}| \deg i)$ for $|\mathcal{D}|$ known destinations. No component needs global topology, synchronized clocks beyond lifetimes, or a learned model shipped from elsewhere.

Evaluation Method

Simulator. We implemented a discrete-event simulator of TRCN in Python, including TDF fragmentation with selective-repeat bitmaps, custody, receipts, strict-priority queues with capacity limits and lower-priority eviction, airtime-accounted beacons broadcast once per interface at the slowest member's LoRa spreading factor, shared collision domains with carrier-sense probability 0.9, scheduled and Markov link outages, node crashes, and blackhole nodes. Sixteen unit tests check the airtime equation against reference values, link-process statistics, collision bookkeeping, posterior updating and forgetting, custody invariants, and STAR's preference for a stable route over a flaky shortcut.

Scenarios. (i) *Seven-node path*: a replication of the original TRCN benchmark, with a single cross-region backbone that returns 80 s after 1–80 messages are queued. (ii) *Two-region disaster network*: 37 nodes per seed (two regions, each with a gateway, a coordinator, four shelters placed at random in an 8×8 km area, and three phones per shelter, plus a hill-top relay). Shelter and gateway LoRa links use SF7–SF11 by distance; 35% are *flaky* (mean up and down periods 90–240 s) and the rest stable (up 1800–3600 s, down 20–60 s). Phones attach to their shelter over Wi-Fi Aware with intermittent availability, sometimes to a second shelter, and to each other over BLE. The regions connect by a fast but intermittent microwave link, a slow and stable LoRa relay, and, when near enough, a flaky border link. Each phone generates Poisson traffic (default 4 messages per hour; 10% SOS to a coordinator; 70% of private messages cross regions); shelters send supply reports. Traffic runs for 3,600 s after a 900 s warm-up. (iii) *Aftershock*: at mid-run the microwave backbone and 30% of shelter LoRa links fail permanently and one shel-

ter crashes for 10 minutes. (iv) *Load sweep*: 1–16 messages per phone per hour. (v) *Route poisoning*: $k \in \{0, 1, 2, 3\}$ shelters advertise perfect routes (with the latest real sequence numbers, the strongest attack that authentication allows) and drop everything they accept.

Baselines. *Static*: the v3.2 behavior, with shortest next hops precomputed from nominal link parameters and never updated. *ETX-DV*: a distance-vector protocol with an expected-transmission-time metric (De Couto et al., 2003) and the same Babel feasibility and poisoned reverse as STAR, hence a strong, loop-free mesh baseline. *Q-routing* (Boyan & Littman, 1994) with ϵ-greedy exploration. *PRoPHET* (Lindgren et al., 2003) with its published parameters, *Epidemic* (Vahdat & Becker, 2000), and binary *Spray-and-Wait* with $L = 6$ (Spyropoulos et al., 2005); these three are multi-copy protocols with summary vectors. All adaptive baselines use the same beacon period and accounted airtime.

Metrics. Packet delivery ratio (PDR, delivered within lifetime), confirmed ratio (receipt returned), SOS PDR, median and 95th-percentile latency of delivered messages, and total airtime (data, acknowledgments, and control) per delivered message. To assess prediction we record Ψ for each message's first decision at its source and score it with the Brier score (Brier, 1950; Gneiting & Raftery, 2007) and the area under the ROC curve (AUC).

Protocol. Parameters were developed on seeds 0–7 and then frozen; all reported results use evaluation seeds 100–119 (10 seeds for load, attack, and path experiments), 1,520 runs in total. A seed fixes topology and traffic, so routers are compared on identical inputs; we report means with 95% t-intervals and paired t-tests across seeds.

Results

Replication of the seven-node benchmark

Table 1 compares our simulator with the original benchmark. Both show every message waiting for the backbone and delay rising with load; at 80 messages the LoRa hops saturate within 600 s. Absolute delays differ because the airtime models differ. On a single path routing has no choices to make, so STAR can only add overhead: its beacons compete with data on the saturated LoRa hops, and it delivers fewer messages than Static within the window at 80 messages (55.9 vs. 64.0). This is the expected cost of a control plane, and it reappears at saturation below.

Table 1

Seven-node path replication. v3.2 = the original TRCN benchmark (single run, bit-rate airtime model); other columns are this simulator (LoRa time-on-air, 10 seeds, means). Deliveries counted within 600 s of submission.

	v3.2 benchmark		Static (this work)		STAR	
Msgs	Deliv.	Delay (s)	Deliv.	Delay (s)	Deliv.	Delay (s)
1	1	84.1	1.0	105.2	1.0	100.8
5	5	97.9	5.0	94.8	5.0	122.7
10	10	114.9	10.0	107.1	10.0	147.2
20	20	149.7	20.0	137.4	20.0	194.8
40	40	222.5	40.0	183.0	40.0	279.8
80	73	339.8	64.0	269.9	55.9	337.5

Baseline disaster scenario

STAR achieved the highest delivery and confirmation ratios and delivered every SOS message (Table 2). Relative to the loop-free ETX-DV baseline, its PDR was 8.0 points higher (paired 95% CI ±2.4, $p < 10^{-5}$) and its median latency 58 s lower (74 vs. 132 s, $p < 10^{-5}$). Relative to PRoPHET, PDR was 7.2 points higher ($p < 10^{-4}$) and median latency 200 s lower, at 32% less airtime per delivery. STAR spent about 19% more airtime per delivered message than ETX-DV. The difference is data airtime, not control airtime (control cost per delivery was nearly equal, 38.7 vs. 38.4 s), which is consistent with the longer but steadier routes that the stability score selects (see the ablation below); Static used the least airtime because it sends no control traffic, but it delivered only 70.9% because it cannot route around outages. Spray-and-Wait performed poorly because shelters do not move: once its copies are spread, it waits for direct contact with the destination, which rarely comes. Its low median latency describes only the few local messages it delivered.

Table 2

Baseline disaster scenario (rate 4 msgs/phone/h; 20 evaluation seeds; mean ± 95% CI). Best value per column in bold.

Router	PDR (%)	Confirmed (%)	SOS PDR (%)	Median lat. (s)	P95 lat. (s)	Airtime/deliv. (s)
STAR	**97.7 ± 0.9**	**95.9 ± 1.3**	**100.0 ± 0.0**	74 ± 11	530 ± 47	74.7 ± 5.0
ETX-DV	89.6 ± 2.5	82.8 ± 3.9	96.7 ± 3.2	132 ± 19	699 ± 25	62.9 ± 4.3
PRoPHET	90.5 ± 2.3	86.5 ± 3.0	98.9 ± 1.2	274 ± 28	802 ± 36	109.8 ± 3.7
Static	70.9 ± 5.2	63.6 ± 5.8	82.1 ± 5.1	257 ± 47	815 ± 26	**16.4 ± 3.5**
Epidemic	75.0 ± 3.1	63.2 ± 3.9	97.4 ± 2.3	399 ± 27	870 ± 18	159.4 ± 4.4
Q-routing	57.4 ± 3.5	37.8 ± 4.0	86.2 ± 5.4	135 ± 31	777 ± 34	89.2 ± 10.5
Spray&Wait	20.9 ± 1.9	16.1 ± 1.5	55.0 ± 7.3	**5 ± 1**	**340 ± 124**	138.2 ± 14.8

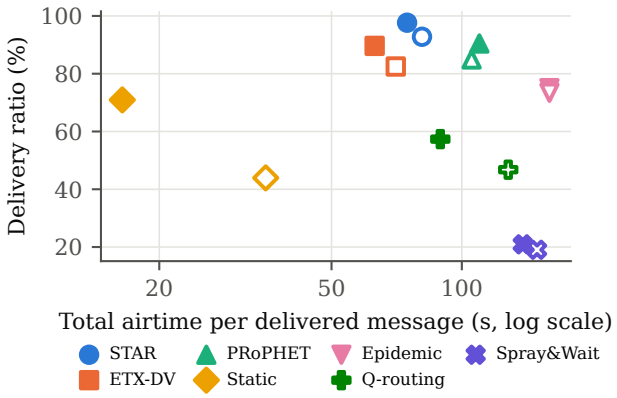


Figure 2

Delivery vs. total airtime per delivered message (log scale). Filled markers: baseline scenario; hollow markers: aftershock (20 seeds, means). STAR sits at the high-delivery end of the frontier at moderate cost.

Load

Figure 3 shows the regime in which learning helps. At low load STAR delivered 99.5% with a 29 s median latency. Its advantage over ETX-DV narrowed with load and disappeared at 12–16 messages per phone per hour (86.1% vs. 85.2% and 78.8% vs. 77.5%), where the LoRa channel, not route choice, limits throughput. Epidemic and PRoPHET, which match STAR at the lowest load, fell to 29% and 45% at the highest load as replicas consumed the channel.

Aftershock

When the fast backbone and 30% of shelter links disappeared, STAR kept the highest PDR (92.8%, Table 3), 10.3 points above ETX-DV ($p < 10^{-7}$) and 7.8 above PRoPHET ($p < 10^{-5}$). Static collapsed to 43.9% because its preloaded routes kept pointing at dead links. Figure 4 shows the dynamics: every protocol loses messages created just before the event, whose routes vanish mid-flight; STAR recovers within one to two bins as its availability posteriors and feasible distances adapt. PRoPHET delivered slightly more SOS messages after the aftershock (97.1% vs. 95.1%; the intervals overlap) because it replicates everything.

Route poisoning

A blackhole that advertises perfect routes captured almost all traffic of the unprotected ETX-DV protocol: its PDR fell from 88.9% to 9.7% with one compromised shelter (Figure 5). STAR retained 64.2%, 42.0%, and 28.3% with one, two, and three compromised shelters, against 46.7%, 19.5%, and 10.8% for STAR without outcome learning; removing only the watchdog gave 51.0%, 23.2%, and 14.4%. These defenses do not make single-copy routing as robust as replication: PRoPHET and Epidemic retained 50–70%. Part of the gap is structural. With two compromised shelters, only about 70% of messages in our topologies have any attacker-free path, because some phones attach only to a compromised shelter. The remainder reflects the learning delay before trust falls below the exclusion threshold.

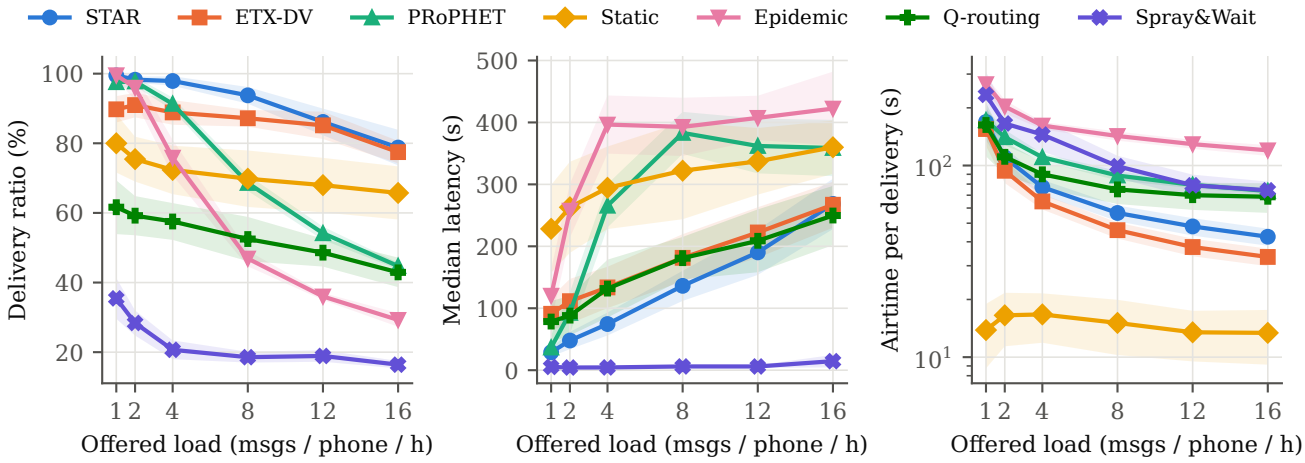


Figure 3

Load sweep (10 seeds per point; bands are 95% CIs). STAR leads up to 8 messages per phone per hour; at saturation (12–16) its delivery converges with ETX-DV, while the replication protocols collapse as copies exhaust LoRa airtime.

Table 3

Aftershock scenario: at mid-run the microwave backbone and 30% of shelter LoRa links fail permanently and one shelter crashes for 10 min (20 seeds).

Router	PDR (%)	Confirmed (%)	SOS PDR (%)	Median lat. (s)	P95 lat. (s)	Airtime/deliv. (s)
STAR	**92.8 ± 2.3**	**87.9 ± 2.9**	95.1 ± 3.6	97 ± 18	634 ± 52	80.9 ± 4.0
ETX-DV	82.5 ± 2.5	71.5 ± 3.5	91.0 ± 4.4	172 ± 26	744 ± 29	70.4 ± 4.4
PRoPHET	84.9 ± 3.0	79.7 ± 4.3	**97.1 ± 3.2**	242 ± 29	766 ± 40	105.3 ± 3.8
Static	43.9 ± 3.8	39.2 ± 3.4	59.6 ± 11.5	199 ± 39	786 ± 45	**35.2 ± 10.4**
Epidemic	73.4 ± 3.2	62.4 ± 4.2	96.0 ± 3.1	410 ± 33	864 ± 19	159.4 ± 4.3
Q-routing	46.8 ± 2.6	27.7 ± 3.3	77.8 ± 5.7	170 ± 41	792 ± 41	128.0 ± 10.4
Spray&Wait	19.1 ± 1.6	15.1 ± 1.3	46.2 ± 7.4	**6 ± 2**	**463 ± 120**	149.2 ± 15.6

Ablation

Table 4 isolates the components. The stability score is the component with the largest and only statistically significant effect: replacing Ψ by minimum expected delay lowered PDR by 2.3 points in the baseline scenario ($p = 0.002$) and by 4.2 points after the aftershock ($p < 10^{-4}$), while reducing median latency by 11–19 s. This is the expected trade-off: stability-aware choice accepts slightly longer routes that are less likely to fail. The other components had no detectable effect on delivery in attack-free scenarios. Thompson sampling, discounting, SOS replication, the watchdog, and end-to-end learning changed PDR by less than 1 point (all $p > 0.1$), and the security mechanisms cost about 9–12 s of median latency because their conservative trust estimates occasionally steer traffic away from honest neighbors. Their value appears under attack (Figure 5) and, for discounting and exploration, in non-stationary conditions that our 1-hour scenarios exercise only partly. We report these null results because they bound what the full model should be credited with.

Prediction quality

The raw score is informative but conservative (Figure 6). Over 4,408 source decisions it separated delivered from lost messages with AUC 0.69, but its

Table 4

Ablation of STAR components (20 evaluation seeds, mean ± 95% CI). PDR = packet delivery ratio.

		Baseline scenario			Aftershock scenario		
Variant	Removed component	PDR (%)	SOS (%)	Med. lat. (s)	PDR (%)	SOS (%)	Med. lat. (s)
STAR	Full model	97.7 ± 0.9	100.0 ± 0.0	74 ± 11	92.8 ± 2.3	95.1 ± 3.6	97 ± 18
STAR-noTS	no Thompson sampling (posterior means)	97.8 ± 0.7	99.2 ± 1.1	80 ± 11	92.5 ± 2.6	97.9 ± 1.7	108 ± 23
STAR-noRisk	no risk-averse LCB for P0/P1	97.6 ± 0.9	100.0 ± 0.0	74 ± 10	92.4 ± 2.1	95.9 ± 3.6	99 ± 18
STAR-noStab	no stability score ($\Psi \to$ min. delay)	95.4 ± 1.5	99.2 ± 1.1	63 ± 10	88.6 ± 2.9	97.2 ± 2.8	78 ± 13
STAR-noE2E	no end-to-end learning or trust	98.2 ± 1.0	99.5 ± 1.0	64 ± 11	92.5 ± 2.6	98.7 ± 1.9	85 ± 19
STAR-noWatch	no watchdog overhearing	98.3 ± 0.7	100.0 ± 0.0	65 ± 10	92.8 ± 2.6	97.9 ± 2.2	93 ± 18
STAR-noSpray	no SOS replication	98.1 ± 0.8	100.0 ± 0.0	72 ± 11	92.5 ± 2.1	95.4 ± 3.6	102 ± 20
STAR-noDecay	no posterior discounting	98.3 ± 0.6	100.0 ± 0.0	69 ± 9	92.5 ± 2.5	98.8 ± 1.8	93 ± 15

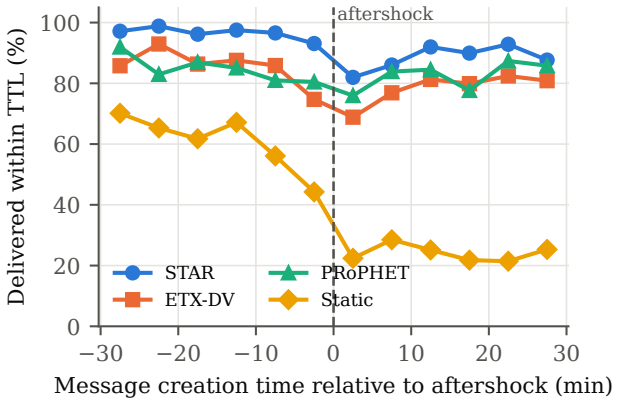


Figure 4

Delivery ratio by message creation time around the aftershock (20 seeds pooled, 5-minute bins).

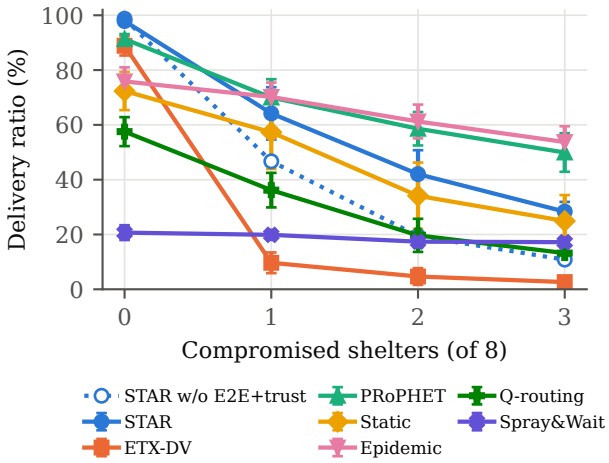


Figure 5

Delivery under blackhole shelters that advertise perfect routes and drop traffic (10 seeds; error bars are 95% CIs). The dotted line is STAR without end-to-end learning, trust, or watchdog.

mean (0.85) was below the observed delivery rate (0.96), so its Brier score (0.069) was worse than a constant forecast at the base rate (0.038). The bias has a clear cause: Ψ evaluates a *fixed* route, whereas downstream nodes re-route adaptively, and messages that miss one opportunity usually find another. Ψ therefore behaves as a lower bound, which is the safe direction for a quantity that triggers SOS replication. Because the bias is monotone, a two-parameter logistic recalibration learned from receipt outcomes fixes it: fitted on seeds 100–109 and tested on 110–119, the recalibrated score had a Brier score of 0.038 against 0.040 for the base-rate forecast, with predicted and observed frequencies within 1–2 points in the bins that hold 97% of messages. Ranking, which drives routing, is unaffected by such recalibration; reported delivery probabilities, for example in a user interface, should use the recalibrated value.

Discussion

What the model learns. STAR's gains come from two sources. First, availability posteriors with silence-aware decay let a node distinguish a lossy-but-present link from an absent one and hold traffic rather than retransmit into a dead link, which matters on LoRa, where a failed exchange can waste seconds of channel time. Second, the stability score prefers routes whose delay is both short and predictable relative to the remaining lifetime.

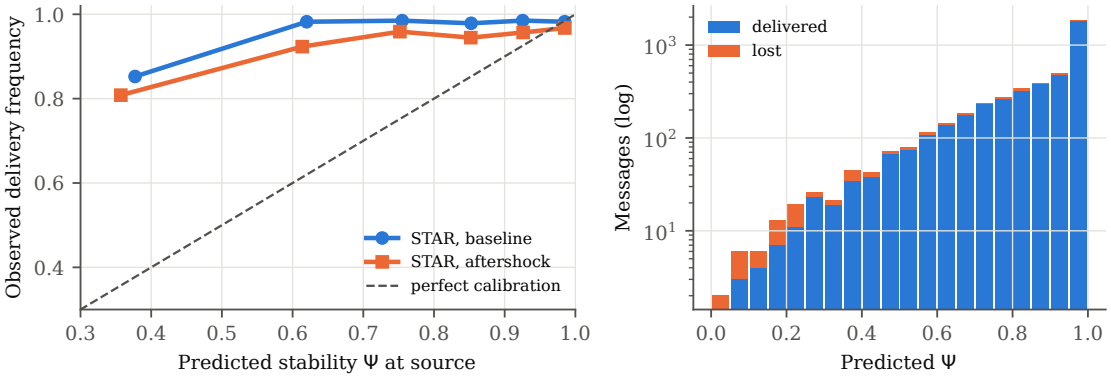


Figure 6

Left: reliability of the raw stability score Ψ *at the source (bins with at least 10 messages). Right: distribution of* Ψ *for delivered and lost messages (log scale).*

Shortest-delay routing gets the first benefit but not the second, which explains the ablation.

When not to expect gains. At saturation, delivery is bounded by channel capacity, and STAR's slightly larger control entries and longer routes cost about as much as better route choice gains. On single paths there is nothing to choose. Against insiders, single-copy custody is fundamentally exposed while trust is being learned; a deployment could replicate SOS messages unconditionally during the first minutes after a topology change.

Privacy. Advertisements expose which mailboxes are reachable through which neighbors. TRCN's rotating pairwise mailboxes limit linkage across epochs, but routing summaries remain metadata that a passive observer can collect; aggregating destinations by region would reduce exposure at the cost of precision.

Safety. STAR never reports delivery itself. Only signed receipts do, consistent with TRCN's rule that the user interface must never show a false success. Predicted probabilities shown to users should be recalibrated, and phrased as estimates.

Threats to validity. All results are synthetic. The link process is exponential on/off with Bernoulli frame loss, whereas real outages are often heavy-tailed and spatially correlated. Collision domains are region-wide with no capture effect, hidden terminals, or half-duplex constraints. Duty-cycle limits and Taiwanese low-power radio regulations (National Communications Commission, 2024) are not modeled, phones do not move between shelters, and the blackhole attacker is not adaptive (it does not, for example, forward only overheard traffic). Parameters were tuned on development seeds of the same generator, so results on other topologies may differ. Each of these is a direction for field measurement, not a claim of the paper.

Conclusion

STAR shows that a node can predict, using only what it observes, which neighbor is most likely to deliver a message on time, and can improve that prediction from the outcomes of its own decisions. Composing Bayesian link posteriors with loop-free, authenticated summaries into a deadline-aware stability score raised delivery and lowered latency against strong baselines in synthetic disaster networks, and outcome-calibrated trust kept route poisoning from capturing the network. The next steps are durable custody, real radios under legal operating conditions, and traces from communities that would use such a network.

Data and Code Availability

The simulator (`trcn_star/`), experiment scripts (`experiments/`), raw per-run results (`results/`), and unit tests (`tests/`) are available at https://github.com/chengmatt416/trcn.

`python experiments/run_all.py` followed by `python experiments/plots.py` regenerates every table and figure.

Use of Generative AI

This research was conducted with substantial assistance from generative artificial intelligence. The initial TRCN design notes (protocol specification, threat model, benchmark notes, and roadmap) on which this work builds were produced with ChatGPT (OpenAI). Claude (Anthropic; model Claude Opus 5.5) was then used, under the author's direction, to (a) search the literature and check each reference against publisher, standards-body, DBLP, or arXiv records; (b) design the STAR model and implement the simulator, baselines, unit tests, and experiment scripts; (c) run the experiments and statistical analyses; and (d) draft and edit the manuscript, tables, and figures. All numerical results come from executing the released code, not from text generated by a model, and can be regenerated with the scripts listed under Data and Code Availability. The author, Pinyen Cheng, defined the research goal, reviewed the outputs, and takes full responsibility for the content of this article.

References

Adelantado, F., Vilajosana, X., Tuset-Peiro, P., Martinez, B., Melia-Segui, J., & Watteyne, T. (2017). Understanding the limits of LoRaWAN. *IEEE Communications Magazine*, *55*(9), 34–40. https://doi.org/10.1109/MCOM.2017.1600613

Augustin, A., Yi, J., Clausen, T., & Townsley, W. M. (2016). A study of LoRa: Long range & low power networks for the internet of things. *Sensors*, *16*(9), 1466. https://doi.org/10.3390/s16091466

Baccour, N., Koubâa, A., Mottola, L., Zúñiga, M. A., Youssef, H., Boano, C. A., & Alves, M. (2012). Radio link quality estimation in wireless sensor networks: A survey. *ACM Transactions on Sensor Networks*, *8*(4), Article 34. https://doi.org/10.1145/2240116.2240123

Balasubramanian, A., Levine, B., & Venkataramani, A. (2007). DTN routing as a resource allocation problem. *Proceedings of the 2007 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communications (SIGCOMM '07)*, 373–384. https://doi.org/10.1145/1282380.1282422

Bhavani, A., Ramana, A. V., & Chakravarthy, A. S. N. (2023). A review on machine learning based routing protocols for delay tolerant networks. *Intelligent Decision Technologies*, *17*(2), 287–299. https://doi.org/10.3233/IDT-220018

Bor, M., Roedig, U., Voigt, T., & Alonso, J. M. (2016). Do LoRa low-power wide-area networks scale? *Proceedings of the 19th ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM '16)*, 59–67. https://doi.org/10.1145/2988287.2989163

Boyan, J. A., & Littman, M. L. (1994). Packet routing in dynamically changing networks: A reinforcement learning approach. *Advances in Neural Information Processing Systems 6*, 671–678.

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, *78*(1), 1–3. https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

Burgess, J., Gallagher, B., Jensen, D., & Levine, B. N. (2006). MaxProp: Routing for vehicle-based disruption-tolerant networks. *Proceedings of IEEE INFOCOM 2006*. https://doi.org/10.1109/INFOCOM.2006.228

Central News Agency. (2024, April 3). *強震影響三大電信 172 個基地台 NCC：估今天可修復 [Strong earthquake affects 172 base stations of the three major carriers; NCC expects repairs today]*. Retrieved November 8, 2026, from https://www.cna.com.tw/news/ahel/202404030090.aspx

Choi, S. P. M., & Yeung, D.-Y. (1996). Predictive Q-routing: A memory-based reinforcement learning approach to adaptive traffic con-

trol. *Advances in Neural Information Processing Systems 8*, 945–951.

Chroboczek, J., & Schinazi, D. (2021). *The Babel routing protocol* (Request for Comments No. 8966). Internet Engineering Task Force. https://doi.org/10.17487/RFC8966

De Couto, D. S. J., Aguayo, D., Bicket, J., & Morris, R. (2003). A high-throughput path metric for multi-hop wireless routing. *Proceedings of the 9th Annual International Conference on Mobile Computing and Networking (MobiCom '03)*, 134–146. https://doi.org/10.1145/938985.939000

Dudukovich, R., Hylton, A., & Papachristou, C. (2017). A machine learning concept for DTN routing. *2017 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*, 110–115. https://doi.org/10.1109/WiSEE.2017.8124902

Fall, K. (2003). A delay-tolerant network architecture for challenged internets. *Proceedings of the 2003 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communications (SIGCOMM '03)*, 27–34. https://doi.org/10.1145/863955.863960

Fraire, J. A., De Jonckère, O., & Burleigh, S. C. (2021). Routing in the space internet: A contact graph routing tutorial. *Journal of Network and Computer Applications*, *174*, 102884. https://doi.org/10.1016/j.jnca.2020.102884

Gai, Y., Krishnamachari, B., & Jain, R. (2012). Combinatorial network optimization with unknown variables: Multi-armed bandits with linear rewards and individual observations. *IEEE/ACM Transactions on Networking*, *20*(5), 1466–1478. https://doi.org/10.1109/TNET.2011.2181864

Garivier, A., & Moulines, E. (2011). On upper-confidence bound policies for switching bandit problems. *Algorithmic Learning Theory (ALT 2011)*, *6925*, 174–188. https://doi.org/10.1007/978-3-642-24412-4_16

Gilbert, E. N. (1960). Capacity of a burst-noise channel. *Bell System Technical Journal*, *39*(5), 1253–1265. https://doi.org/10.1002/j.1538-7305.1960.tb03959.x

Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, *102*(477), 359–378. https://doi.org/10.1198/016214506000001437

Guerrero Zapata, M., & Asokan, N. (2002). Securing ad hoc routing protocols. *Proceedings of the 1st ACM Workshop on Wireless Security (WiSe '02)*, 1–10. https://doi.org/10.1145/570681.570682

Jain, S., Fall, K., & Patra, R. (2004). Routing in a delay tolerant network. *Proceedings of the 2004 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communications (SIGCOMM '04)*, 145–158. https://doi.org/10.1145/1015467.1015484

Keränen, A., Ott, J., & Kärkkäinen, T. (2009). The ONE simulator for DTN protocol evaluation. *Proceedings of the 2nd International Conference on Simulation Tools and Techniques (SIMUTools '09)*. https://doi.org/10.4108/ICST.SIMUTOOLS2009.5674

Khamaisi, K., Kamer, O., Rodrigues, B., von der Assen, J., & Stiller, B. (2025). Bridging technical capability and user accessibility: Off-grid civilian emergency communication. *Proceedings of the International Conference on the Internet of Things (IoT 2025)*. https://doi.org/10.1145/3770501.3770505

Kumar, S., & Miikkulainen, R. (1997). Dual reinforcement Q-routing: An on-line adaptive routing algorithm. *Smart Engineering Systems: Neural Networks, Fuzzy Logic, Data Mining, and Evolutionary Programming*, *7*.

Lattimore, T., & Szepesvári, C. (2020). *Bandit algorithms*. Cambridge University Press. https://doi.org/10.1017/9781108571401

Lindgren, A., Doria, A., & Schelén, O. (2003). Probabilistic routing in intermittently connected networks. *ACM SIGMOBILE Mobile Computing and Communications Review*, *7*(3), 19–20. https://doi.org/10.1145/961268.961272

Marti, S., Giuli, T. J., Lai, K., & Baker, M. (2000). Mitigating routing misbehavior in mobile ad hoc networks. *Proceedings of the 6th Annual International Conference on Mobile Computing and Networking (MobiCom '00)*, 255–265. https://doi.org/10.1145/345910.345955

Meshtastic. (2026). *Meshtastic documentation*. Retrieved November 8, 2026, from https://meshtastic.org/docs/

Ministry of Digital Affairs. (2024, April 6). *Ministry of Digital Affairs and TTC transported low-earth orbit satellite equipment to Tianxiang to assist in establishing a disaster relief communication network* [Press release]. Retrieved November 8, 2026, from https://moda.gov.tw/en/press/press-releases/12020

National Communications Commission. (2024). *Low-power radio-frequency devices technical regulations (LP0002)*. Retrieved November 8, 2026, from https://www.ncc.gov.tw/chinese/files/24020/538_49880_240206_3.pdf

Perkins, C. E., & Bhagwat, P. (1994). Highly dynamic destination-sequenced distance-vector routing (DSDV) for mobile computers. *Proceedings of the Conference on Communications Architectures, Protocols and Applications (SIGCOMM '94)*, 234–244. https://doi.org/10.1145/190314.190336

Qvist, M. (2026). *Reticulum network stack*. Retrieved November 8, 2026, from https://reticulum.network/

Raj, V., & Kalyani, S. (2017). Taming non-stationary bandits: A Bayesian approach. https://doi.org/10.48550/arXiv.1707.09727

Reina, D. G., Askalani, M., Toral, S. L., Barrero, F., Asimakopoulou, E., & Bessis, N. (2015). A survey on multihop ad hoc networks for disaster response scenarios. *International Journal of Distributed Sensor Networks*, *11*(10), 647037. https://doi.org/10.1155/2015/647037

Rolla, V. G., & Curado, M. (2013). A reinforcement learning-based routing for delay tolerant networks. *Engineering Applications of Artificial Intelligence*, *26*(10), 2243–2250. https://doi.org/10.1016/j.engappai.2013.07.017

Rusek, K., Suárez-Varela, J., Almasan, P., Barlet-Ros, P., & Cabellos-Aparicio, A. (2020). RouteNet: Leveraging graph neural networks for network modeling and optimization in SDN. *IEEE Journal on Selected Areas in Communications*, *38*(10), 2260–2270. https://doi.org/10.1109/JSAC.2020.3000405

Russo, D. J., Van Roy, B., Kazerouni, A., Osband, I., & Wen, Z. (2018). A tutorial on Thompson sampling. *Foundations and Trends in Machine Learning*, *11*(1), 1–96. https://doi.org/10.1561/2200000070

Semtech Corporation. (2013). *LoRa modem designer's guide (AN1200.13, revision 1)*. Semtech Corporation.

Spyropoulos, T., Psounis, K., & Raghavendra, C. S. (2005). Spray and wait: An efficient routing scheme for intermittently connected mobile networks. *Proceedings of the 2005 ACM SIGCOMM Workshop on Delay-Tolerant Networking (WDTN '05)*, 252–259. https://doi.org/10.1145/1080139.1080143

Talebi, M. S., Zou, Z., Combes, R., Proutiere, A., & Johansson, M. (2018). Stochastic online shortest path routing: The value of feedback. *IEEE Transactions on Automatic Control*, *63*(4), 915–930. https://doi.org/10.1109/TAC.2017.2747409

Thompson, W. R. (1933). On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. *Biometrika*, *25*(3/4), 285–294. https://doi.org/10.2307/2332286

Vahdat, A., & Becker, D. (2000). *Epidemic routing for partially-connected ad hoc networks* (tech. rep. No. CS-2000-06). Duke University.

Appendix
Parameters

Table A1

STAR parameters (frozen before evaluation)

Symbol	Value	Meaning
B	60 s	beacon period (all adaptive routers)
$(\alpha_0,\beta_0)_f$	(4, 1)	frame-success prior
$(\alpha_0,\beta_0)_u$	(3, 1)	availability prior
H_f, H_u	900, 240 s	posterior half-lives
H_s	600 s	staleness half-life of claims
H_e	1200 s	half-life of end-to-end counts
K	4	pseudo-counts for a calibrated claim
τ_{min}	0.4	trust exclusion threshold
ε, ε_{P0}	0.03, 0.01	Ψ tolerance for delay tie-break
Ψ^*, L_{max}	0.95, 3	SOS replication target and cap
z	1.28	lower-bound quantile (10%) for P0/P1
—	24	advertisement entries per beacon
—	4	beacons per sequence-number increment
—	4	rounds per exchange